

Curriculum Vitae - Giovanni Canestrari

31st July 2026

1 Education

- University of Roma Tor Vergata (Nov 2021 - May 2025) *Phd in mathematics*.
Advisor: Carlangelo Liverani.
Thesis: 'Fine properties of correlations in hyperbolic dynamics'.
- University of Bologna (Oct 2018 - Feb 2021) *M.Sc. in theoretical physics, 110/110 cum laude*.
- University of Bologna (Oct 2015 - Dec 2018) *B.Sc. in physics, 110/110 cum laude*.
- Liceo Statale 'L.Galvani' (2010-2015) *Diploma di maturità classica, 100/100*.

2 Employment

- University of Toronto (August 2025 - ongoing) *Walter Craig Postdoc* mentored by Jacopo De Simoi.
- University of Bologna (September 2024 - July 2025) *Assegno di ricerca* mentored by Marco Lenci.

3 Publications

1. Discontinuities cause essential spectrum (with Oliver Butterley and Sakshi Jain), Communications in Mathematical Physics (2023);
2. Discontinuities cause essential spectrum on surfaces (with Oliver Butterley and Roberto Castorrini), Annales Henri Poincaré (2024);

3. Heat equation from a deterministic dynamics (with Carlangelo Liverani and Stefano Olla), *Inventiones mathematicae* (2026).
4. Uniformly global observables for 1D maps with an indifferent fixed point (with Marco Lenci), preprint.
5. On linear response for discontinuous perturbations of smooth endomorphisms, *Advances in Mathematics* (2026).
6. Linear response for Sinai billiards with small holes, preprint.
7. K-mixing for aperiodic Lorentz gases (with Marco Lenci), preprint.

4 Academic Activities

Talks:

1. Dynamic seminar, ICTP, Trieste (28/07/2022).
2. Dynamic seminar, university of Pisa (9/11/2022).
3. Dynamic seminar, university of Bologna (4/5/2023).
4. Workshop on Partial Hyperbolicity, university of Maryland, short talk (8/05/2023).
5. Workshop DAI day, university of Bologna (18/12/2023).
6. DocTorV seminar, university of Rome Tor Vergata (4/05/2024).
7. Workshop on Macroscopic Behavior of Deterministic Systems, university of Maryland (09/09/2024).
8. Dynamics seminar at university of Maryland (03/10/2024).
9. Dynamic seminar at university of Bologna (24/03/2025).
10. Dynamic seminar at university of Lille (28/03/2025).
11. Online DinAmicI seminar (8/04/2025).
12. Workshop Beyond Uniform Hyperbolicity at ICTP Trieste (06/06/2025).
13. School on dynamical billiards at Cortona (29/06/2025).
14. Dynamic seminar at university of Toronto (15/09/2025).
15. Workshop on Dynamical Systems and Mathematical Physics, Great Bay University, Dongguan, China, (19/10/2025).
16. Dynamic seminar at Maryland university (30/04/2026).

Visiting periods:

1. Maryland university May - June 2023.
2. Université Paris Dauphine July 2023.
3. Stony Brook university November 2023.
4. Maryland university September - November 2024.
5. Maryland university April - May 2026.

Service for the mathematical community:

1. Referee and quick opinions for: Communications in mathematical physics, Journal of Dynamical and Control Systems, Journal of Statistical Physics.
2. Reviewer for mathematical reviews of the AMS.
3. Coorganizer of the dynamic seminars at university of Toronto (Fall 2025-ongoing).

Attended Conferences and Schools:

1. 'Modern Aspects of Dynamical Systems', 20-24 July 2020 ;
2. DinAmici Day at Parma, 17/12/2021;
3. Winter school at Grenoble 'Billards dans le Vercors', 17/1/2022-21/12/2021
4. DinAmici Conference at Varese, 6/6/2022-9/6/2022;
5. 'Méthodes probabilistes pour les systèmes dynamiques aléatoires et variant avec le temps' at CIRM, 3/10/2022-7/10/2022;
6. 'Statistical and Computational Aspects of Dynamics' at Centro De Giorgi – SNS, Pisa. December 13 – 16, 2022.
7. 'La dynamique des systèmes d'Anosov', Marseille, 10-14 April 2023.
8. 'Partial Hyperbolicity' in Maryland, 8-14 May 2023.
9. Summer school on 'Partial Hyperbolicity' in Maryland, 1-10 June 2023.
10. 'Microlocal and Probabilistic Methods in Geometry and Dynamics', from 3 to 7 July 2023 (contributed with a poster).
11. 5th DAI day in Bologna, 18/12/2023.
12. School on probabilistic aspects of hyperbolic systems (August 12, 2024 - August 16, 2024), Budapest.

13. Workshop on statistical properties of chaotic dynamics in and out of equilibrium (August 19, 2024 - August 23, 2024), Budapest (contributed with a poster).
14. DinAmicI in Rio, Dynamics, Applications, Interactions (September, 2-6, 2024), Rio de Janeiro (contributed with a poster).
15. Workshop on ‘Macroscopic Behavior of Deterministic Systems’, (09/09/2024) at Maryland University.
16. Workshop on dynamical systems at Penn State university, (14-17/11/2024).
17. Beyond Uniform Hyperbolicity conference (02/06/2025 - 14/06/2025).
18. School on dynamics of billiards at Cortona (29/06/2025- 4/06/2025).
19. Workshop on Dynamical Systems and Mathematical Physics, Great Bay University, Dongguan, China (December 15-19, 2025).

Other:

1. member of the UMI group DinAmici.
2. member of the prin 2022 project ‘Stochastic Properties of Dynamical Systems’.
3. I designed and coded in Julia the animated DAI logo at <https://www.dinamici.org/>.

5 Teaching

Instructor

2026 Introduction to nonlinear dynamics and chaos (MAT332 - University of Toronto).

2025 Introduction to mathematical game theory (MAT406 - University of Toronto).

Teaching assistant

2023 Analysis 2 for engineering sciences (University of Rome Tor Vergata).

2024 Analysis 2 for applied chemistry, data science and material science (University of Rome Tor Vergata).

2025 Analytical mechanics (lagrangian and hamiltonian mechanics) for physicists (University of Bologna).